

SonixED for Nitrate removal



Figure 1: Surface water

According to the 2021 report of the European Commission, at 14,1% of groundwater stations within Europe, Nitrate levels exceed the limit of 50 mg/l as NO₃⁻. An increase to the previous report. High Nitrate levels lead not only to health hazards such as cancer and methemoglobinemia (blue baby syndrome) but have also a big economic impact of up to 320 bn annual costs in the EU+UK alone.

High levels of Nitrates also contribute to eutrophication in surface waters, causing excessive algae growth, oxygen depletion, and harming aquatic ecosystems. For environmental protection, a legislative push is ongoing in the US and the EU. Globally, the WHO has set the limit of Nitrates in potable water to 50 mg/l as NO₃⁻.

HydroVolta, a Belgian SME, stands as a global pioneer in sustainable water recovery solutions, dedicated to addressing the world's water crisis. Their innovative **SonixED** solution marks a new era in Nitrate removal applications, achieving an impressive recovery rate of up to 98%.

At the core of this solution is the patented **SonixED** technology. A seamless integration of Electrodialysis Reversal (EDR) and Ultrasound technologies. EDR facilitates ion separation, while Ultrasound technology prevents fouling, enhances efficiency and reduces maintenance costs. This innovative fusion enables optimal system performance for an extended time period.

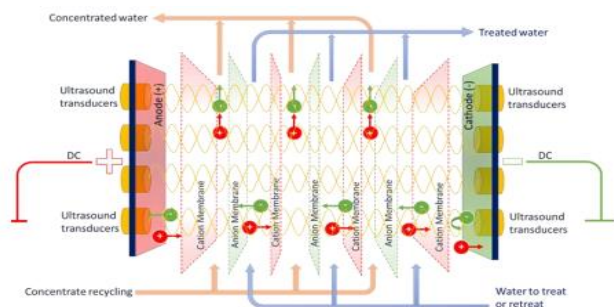


Figure 2: Working principle SonixED

The innovative **SonixED** solution has been advanced further by selectively removing NO₃⁻ ions through electrochemical processes. High-quality electrolytes are created directly from the feed water. This eradicates the need for external chemicals with 95% while simultaneously decreasing maintenance significantly. Tangible benefits include reduced waste, lower CO₂ emissions, and minimized chemical usage. Clearly reflecting in the Total Cost of Ownership (TCO).

This innovation aligns perfectly with the UN SDGs and the European Green Deal aim to move towards zero water waste, preserving groundwater, and reusing water before returning it to the cycle.

The solution is implemented through two standardized container models, that can be used in a scalable way to accommodate the projects needs. Further information and quotations can be requested.



Figure 2: Containerized Ready-To-Use units

Specifications:

	20 ft. container	40 ft. container
Flow rate (per unit)	15-45 m ³ /h	40-120 m ³ /h
Power consumption	0.2 – 0.8 kWh/m ³	0.2-0.8 kWh/m ³
Water recovery	< 98 %	< 98 %
Feed water salinity	< 5.000 ppm TDS	< 5.000 ppm TDS
Product water salinity	< 500 ppm	< 500 ppm
Discharge water salinity	<15.000 ppm	<15.000 ppm
Operating pressure	1-3 bar	1-3 bar
Temperature	5-40°C	5-40°C
pH	2-12	2-12
Free chlorine tolerance	0,5 – 1 ppm	0,5 – 1 ppm

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